A 动态系统

$$\beta \mathbb{E}_t \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} = \frac{1}{\hat{R}_t}$$
$$N_t^{\varphi} C_t^{\sigma} = \hat{W}_t$$
$$\hat{W}_t = \mu A_t$$
$$Y_t = A_t N_t$$
$$Y_t = C_t$$
$$\ln A_t - \ln A = \rho(\ln A_{t-1} - \ln A) + \eta_t^a$$

B 求解模型稳态

B.1 稳态系统

$$\beta = \frac{1}{\hat{R}}$$
$$N^{\varphi} C^{\sigma} = \hat{W}$$
$$\hat{W} = \mu A$$
$$Y = AN$$
$$Y = C$$

B.2 稳态均衡

$$\hat{R} = 1/\beta$$
$$\hat{W} = \mu$$
$$Y = N = C = \mu^{\frac{1}{\sigma+\varphi}}$$

C 对数线性化

$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma}\hat{r}_t$$
$$\varphi n_t + \sigma c_t = \hat{w}_t$$
$$\hat{w}_t = a_t$$
$$y_t = a_t + n_t$$
$$y_t = c_t$$
$$a_t = \rho a_{t-1} + \eta_t^a$$